

Economic and Social Impacts of Entering the Labor Market During a Recession: Evidence from Egypt

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Abstract

1 Introduction

Entering the labor market during a recession has the potential to disrupt or delay the transition to adulthood in a variety of ways. Long unemployment spells or a series of dead-end jobs post-graduation can harm long-run labor market outcomes and may have implications for family formation and health outcomes. Indeed, there is a large literature studying the long-run effects of graduating a recession in high income countries on both labor market and social outcomes. These papers often find that unlucky cohorts have depressed earnings for 5-10 years after graduation, and work in lower-paying firms and in lower paying or poorly matched industries and occupations (XXX citations). Recession entrants in high income countries also fare worse in the marriage market, and have even been shown to die younger (XXX citations). However, despite the fact that searching for one's first job during a recession is challenging regardless of context, we have very limited evidence on recession scarring for labor market entrants in low and middle-income countries. The evidence that does exist is limited in scope and methodological depth due to the challenging data environment, which is a key challenge that I overcome in this paper. ¹

¹The literature on recession entrance in developing countries is considerably smaller. Kuchibhotla, Orazem, and Ravi (2020) and Naidoo, Packard, and Auwalin (2015) examine the effect of unemployment spells at the beginning of an individual's career, but do not use unemployment rate at entry as exogenous variation. Instead of using unemployment rates to as an "instrument" for the probability of being unemployed as a young person, Kuchibhotla, Orazem, and Ravi (2020) use propensity score matching to match similar people where one had a spell of joblessness. Naidoo, Packard, and Auwalin (2015) estimate the effect of early career self-employment on later earnings and use fixed effects to capture unobserved time-invariant worker heterogeneity but otherwise do not have an identification strategy. Cruces, Ham, and Viollaz (2012) is the only paper to my knowledge to use similar methods to the rest of the labor literature to estimate the effect of recession entrance on long-run labor market outcomes. They estimate the effect of entering the labor market in Brazil and Argentina during periods of higher unemployment or higher informality. They only have access to unemployment and informality statistics at the national level, so they use a 2-step process where they estimate cohort effects then regress the cohort effects on the conditions at entry. They find that one percentage point higher unemployment rate lowers the probability of employment by .005 in Argentina and .009 in Brazil over the first 10 years. They only have 2 potential experience bins (0-10 and 11-20) and have no data on job characteristics, so cannot comment on potential mechanisms.

Moreover, there are numerous contextual differences in low- and middle-income labor markets that make extrapolation of existing scarring results from high income settings difficult. Youth unemployment is more common in developing countries, and these youth unemployment spells are often much longer. The impacts of this fact on labor market scarring are unclear: duration effects would suggest scarring may be worse in these settings, but it may also be true that graduating in a recession is easier to recover from when the existing equilibrium has very high rates of joblessness. Additionally, characteristics of developing labor markets such as large informal sectors, large public sectors, and high rates of self employment all suggest that the effect of entering the labor market in a recession in a developing country is an empirical question unanswered by the current literature.

Evidence from high-income countries on family formation impacts of recession entrance is also not very informative. Marriage is a stronger norm in many low- and middle-income countries and is often accompanied by large payments at the time of marriage. Here also the extrapolation of results from high income to low income settings is ambiguous. Would the effects be larger or smaller in a setting where marriage is more ubiquitous? Does the existence of marriage payments help clear the market or add an additional hurdle to scarred cohorts? The importance of marriage in the transition to adulthood and its accompaniment with large cash flows make the effects of graduating during a recession on marriage outcomes, if anything, more economically interesting in a developing setting.

A key barrier to studying recession scarring outside high income countries is the data requirements, which I overcome in this paper. First, sub-national unemployment rate records are needed in order to identify the effect of the unemployment rate at entry while also controlling for cohort effects. Second, one needs a sufficiently long time series of labor market entry for statistical power. There are relatively few low and middle-income countries that began measuring and publishing sub-national unemployment rates early enough to satisfy these requirements. Importantly, Egypt has national statistics records dating to 1968 which record subnational unemployment rates. A major contribution of this paper is the digitization of these records.

Using publicly available labor market panel data alongside the newly hand-digitized historical unemployment rates, I estimate the effect of economic conditions at entry on long-run economic and social outcomes. I find effects which reflect the same kind of economic scarring found in high income countries, but which manifest differently in this setting due to the differences in the “job ladder”. Men do experience reductions in employment for 2 years after entry, but the long run effect on employment is actually positive. However, this increase is mostly driven by an increase in self-employment, and in particular, self employment without any employees. This type of employment is associated with lower earnings, more precarity, and is often understood as disguised unemployment (Majid, 2012).² Male recession entrants are also less likely to have a formal wage work job, which is generally perceived to be the most desirable form of employment due to its stability, access to labor protections, and higher average wages.³

²For example, consider a young person who takes up street vending in order to bring in some income while unable to find wage work. This would be counted as self employment without any employees.

³In the 2018 ELMPS round, formal workers earned on average 29 percent higher wages than informal workers.

Suggestive evidence which examines heterogeneous effects by educational attainment also indicates that scarring effects seem to be largest for college graduates, who are more likely in general to attain formal wage-work jobs and experience the largest declines due to recession entrance. This is consistent with the broader social science literature evaluating the labor market conditions which led to the Egyptian Revolution in 2011, but notably this pattern is also true of prior business cycles in Egypt.

Recession entrance also appears to delay the marriage market significantly, for both men and women. In settings where young people live in their family home until marriage, and where marriage is costly due to brideprice/dowry, high youth unemployment effectively impedes youth people’s transition to adulthood. This stagnation has been dubbed “waithood”, as young people wait to find a job, find a spouse, and for their adult life to begin (Dhillon and Yousef, 2009; Singerman, 2007). This delay is generally perceived quite negatively, evidenced for example by sad anecdotes about strings of broken engagements due to lack of resources (Arraf, 2016) and public information campaigns from Alazhar, a prominent religious institution, promoting modesty and practicality in marriage negotiations instead of excessive displays of wealth (Ramadan, 2020).

Yet despite the undesirable delay to the marriage market in general, there appear to be several silver linings, benefitting women in particular. Women turning 18 during a downturn are more likely to complete secondary school, and the age gap between husbands and wives is slightly reduced for women whose likely marriage cohort is scarred. Perhaps related to more schooling and later marriage, women report higher levels of decisionmaking in the marital home. These results suggest that while policy proposals emphasizing reducing the cost of marriage are important to reducing the barriers young people face to starting families, a balanced approach that does not promote teen marriage may be best.

These results are an important contribution to our understanding of how recessions can affect young people’s long-run outcomes in low and middle income countries. Even (or perhaps especially) in a context marked by high youth unemployment, coming of age during an economic downturn results in permanently lower odds of men reaching the highest rungs of the job ladder. This economic scarring also slows the transition into married life, leaving unlucky cohorts on the fringes of a society where marriage is central. Other work has shown that economic conditions which block men from marriage is a considerable risk factor for civil unrest, emphasizing the importance of taking these scarring effects seriously (Hudson and Matfess, 2017). While Egypt and other countries where groom-to-bride payments are practiced have taken steps to lower the cost of marriage such as facilitating mass weddings, addressing the root issue of protracted joblessness and long term labor market scarring should remain a priority.

The rest of the paper proceeds as follows. Section 2 presents the newly digitized historical unemployment rates and the labor market panel dataset used to measure life outcomes. Section 3 describes the empirical strategy used to estimate the long run effect of recession entrance on labor and marriage outcomes, and Section 4 presents summary statistics before reporting results in Section 5. Section 6 discusses and concludes.

2 Data

To document the long-run impacts of coming of age during a period of high unemployment, several types of data are required. First and most obviously, unemployment rates are needed. National unemployment rates are colinear with cohort effects, so sub-national rates are typically used in this literature. The subsection below describes the digitization of subnational unemployment rates. Second, labor market outcomes are needed, linked to the individuals age and location at labor market entry. This information is gathered from the Egyptian Labor Market Panel Survey (ELMPS) and is described below.

2.1 Unemployment Rates

The Egyptian statistical agency, CAPMAS, reports national unemployment rates in a machine-readable format from 1991 to the present. Governorate-level (the largest administrative division in Egypt) unemployment rates can be calculated from the Labor Force Survey microdata, which is available beginning in 2006. Using cohorts from 2006 and later would significantly limit both the sample size and the extent to which long-run effects may be measured.

CAPMAS also publishes PDF-format statistical bulletins going back to 1968 which contain the number of men and women, by governorate, who are employed, unemployed, or out of the labor force.⁴ With the help of an excellent research assistant, I have digitized these records. The quality of the scan varies from year to year (and page to page), so there are some years in which only 80 percent of records can be transcribed, while some years can be fully transcribed. This scanning difficulty introduces classical measurement error since the quality of a PDF scan is plausibly uncorrelated to the actual unemployment rate. Observations where the digitized employment status statistics do not add up properly to match the reported total (beyond rounding error) due to transcription or other errors are excluded. Several years between 1968-2005 are unavailable on CAPMAS's website, including 1975, 1976, 1985-1988, 1996, and 2003. (Years ending in 6 were census years.) Because the gap in the mid 1980s coincided with a recession, I conduct sensitivity analysis in appendix XXX where I drop all entrants before 1989. The results are largely similar.

To assess the reliability of the statistical bulletins and the transcription of them, figure 7 (XXX rerun this) plots the national unemployment rate as reported by the PDF statistical bulletin alongside the unemployment rate as published in a machine readable format. For years where both statistics are available (1991-present), they are nearly identical. The figure also depicts an upward trend in unemployment rates in the pre-1990 period. This could reflect changes in the survey definition of unemployment over time, although it is also consistent with structural transformation since movement away from agriculture and towards wage work results in higher rates of unemployment.⁵ Because

⁴Records actually go back to 1957, although reports were not conducted annually during this time period, and since the earlier outcome data was collected in 1998, cohorts who entered the labor market prior to 1968 would not contribute meaningfully to the analysis.

⁵Thanks to Julio Garin for pointing this out.

the main specification includes year fixed effects, this is not a threat to identification.

Another important data prerequisite is that the unemployment rate varies sufficiently across governorates. Figure 7 reports residualized gendered unemployment rates, including fixed effects for year, governorate and gender. It is not the case that governorate unemployment rates are level shifts around the national rate, which ensures the data has sufficient variation to identify the parameters of interest.

2.2 Egyptian Labor Market Panel

Outcome data is drawn from the Egyptian Labor Market Panel, a longitudinal labor-focused survey with 4 waves (1998, 2006, 2012 and 2018). The survey is a nationally representative panel survey which tracks households, as well as the new households of individuals who leave households previously in the sample (similar to the PSID) (Krafft, Assad, and Rahman, 2019). Several thousand households are added to each wave as a refresher samples. Table 1 reports the number of individuals in each combination of panel rounds, including all ages. For example, 6,636 individuals appear in the 1998 wave only, 4,143 appear in 1998 and 2006, 3,073 appear in 1998, 2006 and 2012, and 10,145 appear in all 4 waves. Once a household cannot be tracked they are not attempted in the following rounds, so individuals only appear in consecutive waves.

All waves have labor market modules with detailed questions on current labor market participation and job characteristics, plus modules on education, migration, assets, marriage and marriage payments, fertility, and women’s empowerment. Each wave also includes a detailed work history module which asks respondents about their history of jobs and non-wage characteristics of those jobs. These job histories allows for the creation of an annual job panel, which will constitute the main sample of this paper’s labor market analysis. The job history module includes questions about the following characteristics: employment status (wage work/self employment without employees/self employment with employees), sector of employment, permanent/temporary job, whether the job had a contract, whether the job had social security, self-reported firm size, job tenure, industry, and occupation. Because the job history does not have recalled wages or earnings (presumably due to poor recall potential), these outcomes are only measured at the time of the survey.

Job spells are technically reported at the month level, but I aggregate to the year level to focus on longer-term dynamics and because start and end months of jobs are likely subject to more recall error than the years of a job spell. Indeed, over half of job spells reported in the 2018 wave include a year but are missing the month (respondent replied “don’t know”). This rate is even higher in previous waves. When assigning the employment status for a given year, the most chronologically recent status in that year prevails, which will downwardly bias scarring. For example, if an individual was unemployed from 2014-2015, had their first job spell from 2015-2017, followed by a spell of nonemployment from 2017-2018, 2014 would be marked as nonemployed, 2015-2016 employed, and 2017-2018 nonemployed.

Job characteristics are aggregated to the year level following a similar principle. For the 1998, 2006, and 2012 rounds, job histories were collected in a consecutive manner so there are no overlapping jobs in the data. Job characteristics in a particular year are given by the job characteristic of the most chronologically recent job held that year, conditional on being marked as employed that year according to the decision rule described above. The 2018 round allows for holding multiple jobs at once, and approximately 50 percent of job spells have months. Years where spells contain months are assigned according to whichever job was held longer, deferring to the most recent one in the event of ties. Years with job spells without months are assigned the characteristics of most chronologically recent spell. Years with multiple spells overlapping are assigned the “best” characteristic of the concurrent spells. “Best” is assigned as follows, based on associations with higher earnings: public > private, formal > informal, wage work > self employment, permanent > temporary, large firm > small firm, and so forth.

Detailed marriage outcomes are measured in the 2006, 2012 and 2018 ELMPS waves, and the earliest wave with marriage payment data is kept for each individual, following the rationale that recall quality declines with time. Each of these rounds measure the amount in Egyptian Pounds of the mahr and deferred mahr (financially similar to “brideprice” and a “deferred brideprice”/alimony, described in section 4), as well as the total value of jewelry presented to the bride from the groom/groom’s family. Also measured is the total value of furniture, appliances and housing purchased for the marital home and the contribution of the bride, bride’s family, groom, and groom’s family to those expenses. All outcomes are inflation adjusted.

A convenient aspect of using recalled job histories from several waves of a panel is that recall accuracy can be evaluated by assessing the concordance between recalled job histories in different waves. For example, is reported 2005 employment status the same in the 2006 and 2012 wave for individuals in both rounds? To this end, Table 2 reports measures of data quality in the ELMPS and the reliability of the retrospective data.⁶ 99 percent of respondents report the same sex each survey round, and 70 percent of individuals report the same year of birth across all survey rounds. The average range of birth years reported by an individual is 0.76, and the median respondent reports the same birth year each round. Reported date of birth might be misreported due to a parent misreporting a child’s year of birth in an early round, or by elderly individuals who are not certain of their date of birth (which is not uncommon among individuals born in this region in the early-to-mid 20th century.) Respondents are less accurate in their recall of their age at first marriage. 37 percent of married respondents reported the same age at first marriage across rounds, and the average range of different ages at first marriage reported by individuals was 1.86 years, which is relatively small. This may have been misreported due to memory recall errors or due to social desirability bias.

To assess job status recall, “share conflicting employment status” reports the share of years where one or more survey rounds disagree on whether the individual was employed or not, restricted to age 16+. (For example, an individual reporting in the 2006 round that they were employed in 2005, but reporting being nonemployed in 2005 in the 2012 round.) To be conservative, *any* conflicting report counts as a conflict, even if the other 2-3 survey rounds report the

⁶The sample sizes in this table are smaller than the total sample since it restricts to individuals who appear in at least two rounds.

same status. When including a status conflict using any wave of data, the share of years with a conflict is 21 percent. When restricting to conflicting statuses between adjacent rounds, the conflict rate drops to 16 percent. The adjacent rate is the most relevant, since the sample is constructed in such a way that information is always taken from the chronologically closest survey. While not perfect, these statistics lend support to the basic reliability of retrospective data generated from the ELMPS.

3 Empirical Strategy

3.1 Labor Market Outcomes

Following the literature, I regress individual-level labor outcomes on the unemployment rate at entry and include time, cohort, potential experience and governorate of birth fixed effects. To allow for cohort fixed effects, the specification uses governorate-level unemployment rates. Therefore, the effect is identified on variation in unemployment rate within governorate. There are 27 governorates, and as I show in section 4, there is significant variation in governorate level unemployment rate. The specification uses governorate of birth to avoid endogeneity from migration, and the unemployment rate at age 18 to abstract from decisions on when to enter the labor market.⁷

The first and more basic specification estimates a single effect of a higher unemployment rate at entry. UR_{cg}^{entry} is the unemployment rate at age 18 in governorate of birth g for cohort c . exp refers to years of potential experience since age 18. Π_e , Ω_t , Δ_c , and Λ_g are fixed effects for potential experience, calendar time, cohort, and governorate of birth. In order to include fixed effects for calendar, time, and experience, one fixed effect is dropped. β_1 in regression 1 can be interpreted as the average effect of entering the labor market with 1 percentage point higher unemployment over the worker's lifetime. Because contemporaneous unemployment rate is not controlled for in this specification, β_1 estimates the average effect of entry conditions given the typical time path of unemployment rates. This specification will be used to study educational attainment and other less time varying outcomes.

$$y_{icetg} = \beta_0 + \beta_1 UR_{cg}^{entry} + \Pi_e + \Omega_t + \Delta_c + \Lambda_g + \epsilon_{icetg} \quad (1)$$

In order to understand how scarring effects evolve over an individual's career, specification 2 estimates the effect of entry conditions at each year of potential experience and is the preferred specification. β_s estimates the effect of one percentage point higher unemployment on outcome y , s years after age 18. The effect at years 10+ of potential experience is aggregated, which allows for identification.

$$y_{icetg} = \beta_0 + \sum_{s=0}^9 \beta_s UR_{cg}^{entry} * \mathbb{1}\{exp = s\} + \beta_{10+} UR_{cg}^{entry} * \mathbb{1}\{exp \geq 10\} + \Pi_e + \Omega_t + \Delta_c + \Lambda_g + \epsilon_{icet} \quad (2)$$

As a robustness check, I also estimate specification 3 which is the same regression as specification 2 but also controls

⁷18 is the median age at which men and women complete secondary school in the 2018 ELMPS.

for contemporaneous unemployment rate in the governorate of birth.

$$y_{icetr} = \beta_0 + \sum_{s=0}^9 \beta_s UR_{cr}^{entry} * \mathbb{1}\{exp = s\} + \beta_{10+} UR_{cr}^{entry} * \mathbb{1}\{exp \geq 10\} + \beta_c UR_{rt}^{contmp} + \Pi_e + \Omega_t + \Delta_c + \Lambda_r + \epsilon_{icet} \quad (3)$$

3.2 Marriage Outcomes

Because marriage is an absorbing state for most individuals in the sample, a different empirical strategy is needed. To measure whether recession entry delays or accelerates marriage, one could regress age at first marriage on unemployment rate at entry using specification 1. However, this outcome is missing for individuals who are still single, which introduces bias. Instead of restricting the sample to older cohorts who have had sufficient time to get married in the data, I use a cox proportional hazards model to estimate how recession entrance effects the hazard rate into marriage. The standard assumption is that the hazard rate, or the probability of an unmarried person at a given age getting married that year, is affected proportionally over time. This would be violated if the marriage rate was lowered at younger ages by recession entrance but then accelerated later in order to catch up. When proportionality is violated, the estimate can be interpreted as the time weighted average of the effect of recession entrance on marriage rates (Stensrud and Hernán, 2020). Therefore, the bias from any plausible violation of proportional hazards would bias the estimates towards null effects.

Age gaps between husbands and wives also necessitates careful consideration of *when* exactly recessions matter for women vs men with regards to family formation. Husbands in this sample are on average 6 years older than their wives. To the extent that men are primary breadwinners and pay the largest financial marriage costs, it is possible that women might be most affected by the recession scarring of their likely marriage cohort. To account for this, analysis of recession scarring on women’s marriage outcomes will include specifications using the unemployment rate at 18 and at age 12, when their “likely marriage cohort” was turning 18.

Other outcomes related to spousal characteristics, marriage cost amounts, and marital decisionmaking will be analyzed using specification 1, restricting to individuals above the 95th percentile of age at first marriage (29 for women and 33 for men).

4 Context and Descriptive Statistics

The Egyptian labor market is marked by moderate rates of educational attainment, high employment rates for men, and low employment rates for women. Table 3 reports descriptive statistics from each wave of the ELMPS

by gender. There have been persistent increases in educational attainment across time, and the gender gap in educational attainment shrunk considerably between 1998 and 2018. In 1998, 54 percent of men and 34 percent of women aged 25-65 had completed preparatory school (ie, grade 9). 47 percent of men and 30 percent of women had completed secondary, and 18 percent of men and 9 percent of women had completed university. By 2018, 66 percent of men and 54 percent of women had completed preparatory, and 19 percent of men and 16 percent of women had completed university.

Across all waves, approximately of 90 percent of men aged 25-65 are employed, while only 20-28 percent of women are employed. However, this statistic conceals considerable heterogeneity by age. High youth unemployment is a key characteristics of the Egyptian labor market. Figure 3 reports the national total unemployment rate and the national youth unemployment rate (ages 15-24). The total unemployment rate ranges between 7 and 13 percent, while the youth unemployment rate ranges between 17 and 34. Figure 4 displays this high youth unemployment at the individual level, as the gap between the year individuals in the 2018 ELMPS exited school and the year they started their first job. The figure shows the long period of “waithood” quite strikingly. Of ever-working men with at least a secondary school education and who last exited schooling between the ages of 14 and 30, 20 percent held a job before they exited school, 40 percent found their first job within 1 year of exiting school, and approximately 35 percent don’t begin their first job between 2-10 years after exiting school. Ever working women with the same qualification have ever longer gaps between school and work; 35 percent start their first job within 1 year of graduation, and approximately 30 percent begin their first job 2-10 years after exiting school. An additional 10 percent don’t begin working for more than 10 years after exiting school.

There has been considerably less change in family formation timelines than educational attainment during this time. The average age at first marriage for married men aged 25-65 has remained fairly constant around 27, while the age at first marriage for women has increased slightly over time. In 1998, the average age at first marriage was 20, while in 2018 it had risen to 21.4

Egyptian marriages are accompanied by various payments and gifts, some of which are customary (such as gold jewelry) and others which are dictated by prenuptial religious marriage contracts. Except for foreigners, all marriages in Egypt are governed by religious family law pertaining to the couple’s religion. I focus on Islamic marriages as only 5 percent of individuals in the ELMPS are Christian. Islamic jurisprudence requires a payment from the groom to the bride known as *mahr*, which is the sole property of the bride. Islamic marriage contracts also stipulate a “deferred” *mahr* which is owed to the bride in the event of her husband’s death, no-fault divorce initiated by the husband, or fault-divorce proven in court. Prior to 2000, no-fault divorce only existed for men and women could only initiate a divorce in the event of abuse, abandonment, unfaithfulness, undisclosed impotence, etc. No-fault divorce for women (*khul’*) was introduced in 2000, but the wife forfeits the deferred *mahr* and must repay the *mahr* if she chooses this type of divorce (Corradini and Buccione, 2023). As such, the deferred *mahr* is similar but not identical

to alimony negotiated in a prenup.

Another customary marital contract unique to Egypt is the *ayma*, or inventory of goods and wealth brought into the marriage by the wife, plus any wedding gifts bestowed by the groom, all of which is agreed to be the property of the wife (El-Kholy, 2000; Rania, 2018). Its value is negotiated up front between the bride and groom and their families, and enumerates what the wife is owed upon death or divorce. Unlike the mahr muakhr, which is enforceable under personal status laws, the *ayma* falls under civil contracts related to safekeeping of property. As a result, these contracts could be easier to enforce in practice. According to El-Kholy, 2000, negotiating an *ayma* is more common among working class couples and is seen as an important form of economic protection for women.

The ELPMS measures these contractual amounts as well as costs associated with jewelry, furniture, *gihaz* (*preparations*), housing, and the wedding itself, plus the portion of these costs borne by the bride or groom’s side. Summary statistics of these costs are reported in Panel B of Table 3. Total costs of marriage declined from 2006 to 2012, then rose again when measured in 2018. *Ayma* amounts has risen, while inflation-adjusted mahr amounts have fallen, consistent with the practice becoming more ceremonial than economically meaningful.

Panel B also reports how other marital characteristics of note have changed over time. The average age gap between husbands and wives is approximately 6 years and has fallen slightly over time. In 2006, approximately half of couples lived alone in the beginning of their marriage, which has risen to 63 percent in 2018. 30 percent of couples are related by blood in some capacity (first cousins or more distant relatives), and this has remained relatively stable over time. Finally, polygamy is rare, with only 1 percent of married households in the ELPMS including more than 1 wife.

5 Results

5.1 Education

Table 4 reports the effect of entry conditions on educational attainment using specification 1. Columns 1 and 2 report the effect of an additional percentage point of unemployment at age 18 on the probability of completing secondary or post-secondary, restricting to individuals age least 25. Column 3 reports the effect on completing university, restricting to those aged 28+. There are no statistically significant effects for men at any level, ruling out any effects associated with “sheltering” from the poor labor market by extending education, or leaving school earlier to supplement household income.

Women, on the other hand, receive more education. Women turning 18 during a recession are 0.81 percentage points more likely to complete secondary school for each additional point in the unemployment rate at entry. The peak to trough of the average business cycle in Egypt is approximately 4 percentage points, meaning that the average recession increased the probability of a woman completing secondary school by 3.24 percentage points. Results

described below point to a delayed marriage market, which may explain the increase in educational attainment. As the age at first marriage for women is around 20, a delayed marriage market may have allowed women to stay in school longer. It is also possible that the recession influenced women to seek more education with higher labor market returns, and were less burdened by immediate household financial needs than similarly aged men.

5.2 Employment

With null effects on educational attainment, the effects of a recession on men’s employment outcomes can be straightforwardly interpreted. Table 5 reports estimates from specification 2, which regresses annual job outcomes on the governorate of birth unemployment rate at entry interacted with years of potential experience dummies, plus fixed effects for cohort, year, and governorate of birth. The coefficient on each potential experience dummy interacted with the unemployment rate at entry can be interpreted as the impact of an additional percentage point of unemployment at entry, at x years of potential experience. These coefficients allow us to trace the long run effect over time. Column 1 reports the effects on an indicator for having any job, including self employment. There are statistically significant negative effects on the probability of employment for two years following entry. An additional percentage point of unemployment at entry lowers the probability of employment that year by 0.9 percentage points (almost 1:1), the following year by 0.6 percentage points, and the year after by 0.3 percentage points. By six year, there is actually a statistically significant and *positive* effect on the probability of employment that grows over time. In the long run (10+ years after entry), recession entrants are 0.5 percentage points *more* likely to be employed for each additional percentage point of unemployment.

This positive long-run result is somewhat counterintuitive, and none of the recent papers on recession scarring in high-income countries find positive long-run effects. However, this result masks important heterogeneity on the type of employment. Column 2 reports impacts on the probability of being employed in wage work (with anything else, including self employment or non-employment marked as zero). Recession entrants are less likely to be engaged in wage work for 3 years after entry, and there is no long-run positive impact. Instead, recession entrants shift towards self employment. By four years after entry, there is a statistically significant increase in the probability of being self employed, which grows over time and remains significant in the long run. At 10+ years after entry, recession entrants are more likely to be self-employed by 0.3 percentage points for each additional percentage point in the unemployment rate. Multiplied by four, this is associated with a total effect of 1.4 percentage points more likely to be self employed in the long run.

Some types of self-employment are associated with more income than others. Specifically, owning a small business with employees is very different from the casual self-employment that many turn to to match ends meet, such as street vending, odd jobs, etc. Column 4 estimates recession entry effects over time on the probability of being self-employed without any employees. Almost all of the long-run effect on self employment is concentrated in this more

casual form of self-employment, associated with more precarity and lower earnings. These effects are consistent with long-term discouragement from the market for wage work.

Male recession entrants are also less likely to hold formal wage work in the medium- to long-term, which provides additional evidence of slowed job ladder growth. The negative effect becomes statistically significant four years after entry, and only converges to zero in the 10+ year long-run. The fact that this effect only emerges 4 years after age 18 suggests it may be concentrated among the college educated population (who are also more likely to hold formal jobs). This question is investigated in suggestive heterogeneity analysis below.

There are almost no employment recession scarring effects for women, which is perhaps not unexpected given the low rates of labor force participation. With the increase in educational attainment, it is also possible that the null employment impacts are the net result of recession scarring plus a boost from increased educational attainment. The only statistically significant impact reported in Table 6 is a long-run positive impact on the probability of holding a formal waged job. 10+ years after entry, women entering during a recession are 1.7 percentage points more likely to hold a formal waged job. The two most common formal jobs for women in the 2018 ELMPs are administrative assistant and elementary school teacher. One plausible narrative consistent with the long-run positive effect on formal wage work is as follows. With increased education, a mother in her 30s with school-aged children might be more likely to get a teaching or administrative job in a school. However, this is a single estimate among many nulls, so overinterpretation should be avoided.

5.2.1 Heterogeneity by Educational Attainment

This section examines heterogeneity by educational attainment in the scarring effects on men. Because recession entrance had no statistically significant impacts on men's educational attainment at any level, recession-induced changes in who selects into higher educational is likely to be minimal. Nonetheless, these results should be viewed as suggestive.

Tables 12 and 13 report scarring impacts on men attaining secondary school and university, respectively. Both tables report largely null impacts on employment, but relatively large negative impacts on the probability of having a formal job that remain significant in the long run. 10+ years after age 22, college grads remain 0.9 percentage points less likely to have a formal job for each percentage point increase in the unemployment rate at entry. Effects are slightly smaller but still large for secondary grads - 10+ years after age 18, secondary grads are 0.6 percentage points less likely to have a formal job.

Table 11 examines the impact of recession entry on men who did not complete secondary school. There are no statistically significant impacts on employment or wage work vs self employment, but column 4 reports statistically significant increases in the probability of having a formal wage-work job. This is somewhat unexpected, although column 5 clarifies that much of this impact is concentrated among public enterprises (which is a separate sector

code from “government” work). This sector categorization likely includes state enterprises and active labor market programs like the Public Works Programme (Semlali and Angel-Urdinola, 2012; for Development, 1999). Only 4 percent of male, formal wage workers with less than a secondary education report this sector of employment in the 2018 wave, while 47 percent of these workers are in the government sector. This result suggests that public enterprises play a non-trivial role in bolstering access to social security and the safety net for lower skill workers.

5.3 Wages and Earnings

The forgoing results suggest large earnings losses, especially for workers with at least a secondary degree. Unfortunately, there are significantly fewer observations with actual wage or earnings information since these outcomes are only measured at the time of the survey and are excluded from the survey’s job history section. To maximize power and focus on aggregate impacts, table 7 reports results from regressions of earnings or wages, measured in one of the 4 survey rounds, on the unemployment rate at entry using 1. These specifications reveal the aggregate impact across the whole career. Column 1 examines the probability of having earnings below the (male/female) median. Columns 2 and 3 measure scarring effects on earnings below the 75th and 90th percentile, and column 4 reports effects on log hourly wages. Column 1, row 1 shows a statistically significant increase the probability of having below median earnings for men who enter during recessions. Scaling by the average 4 point increase in the unemployment rate during a recession, a recession entrant is, on average across the entire career, 1.3 percentage points more likely to have below median earnings. This effect appears to be stronger for college grads, for whom the effect is larger and statistically significant at the 75 and 90th percentiles as well. The average recession college grad is less likely by 2.6 percentage points to have earnings above the 90th percentile.

5.4 Marriage Market Outcomes

Table 9 shows that recessions significantly delay marriage for both men and women. The table reports hazard ratios which reflect the impact of treatment on the relative hazard rates. Column 1 reports the effect of unemployment rate at age 18 on the time to marriage for men. The point estimate implies that each additional percentage point in the unemployment rate at age 18 lowers the probability of marriage in a given year by 1.5 percent for men. To more concretely contextualize how this impacts marriage timelines, we can consult the survivor function which results from this model. According to the survivor functions, 30.8 percent of 25 year old men at the 25th percentile of entry unemployment rates are married, compared to 28.7 percent of 25 year old men at the 75th percentile. At 30 years old, the gap widens slightly: 70.8 percent of men are married at the 25th percentile of entry conditions vs 67.6 percent of men at the 75th percentile. These magnitudes are meaningful, as they represent thousands of young men for whom the entry into socially sanctioned intimate relationships and the cultural signifier of adulthood is delayed.

The impact of recession entrance has a similar magnitude of impact among women. Columns 3 and 4 reports

the effect of the unemployment rate at age 18 and age 12 (the year that the “likely marriage cohort” turned 18). Both point estimates are statistically significant, and correspond to a 1.1-1.3 percent decrease in the probability of getting married each year. The fact that both estimates are significant suggests that women are affected by recession scarring on the cohort of men they will marry and by recessions at the time of their own labor market/marriage market entry. In other words, recessions can delay marriage in two ways: first, by lowering men’s early career labor market earnings, and by creating financial strain (for everyone) around the time of marriage. Column 2 reports the effect of the unemployment rate at age 12 for men as a placebo test. Reassuringly, there is no impact.

Table 9 reports estimates on the effect of recession entrance on spousal characteristics and marriage costs. Each row/column corresponds to a separate regression. For individuals in multiple rounds, the earliest observation is kept to reduce recall bias. There are no statistically significant effects for men, although the sample sizes for these regressions are smaller since men were not surveyed on the costs of marriage until the 2012 round.

For women, increased unemployment rates at age 18 do not impact spousal characteristics but do increase total groom spending, with much of the increase coming from increased spending on home furnishings, appliances etc. As the price of gold often increases dramatically during times of inflation, there is anecdotal evidence that grooms shift spending toward home furnishings and appliances in lieu of inaffordably expensive gold (Kholaif and El-Ghobashy, 2016). It is also possible families demand more appliances up front instead of buying later to protect their children, in light of tougher economic circumstances (Arraf, 2016).

The unemployment rate at age 12 is also associated with increased spending on home furnishings, and a relatively small but significant reduction in the average age gap between husbands and wives. This effect on the age gap could be caused by women waiting slightly longer than men as they wait for their marriageable cohort to accumulate the funds for marriage. Table 10 also reports that an increase in the unemployment rate at age 12 increases women’s decisionmaking power in their eventual marriages. This could be due to the increased age at first marriage or increases in educational attainment shown above.

6 Conclusion

Despite the severity of youth unemployment in many developing countries, there has been little work to date on the effect of entering the labor market during a recession on long run outcomes in low and middle income countries. There is also virtually no work on how recession entrance affects family formation in cultural settings practicing marriage payments. I fill this gap by digitizing historical Egyptian unemployment rates and estimating scarring effect on both labor and marriage market outcomes. I find scarring effects on the male employment rate which persist for approximately 5 years before converging to zero. Male recession entrants who are employed are less likely to work in the public sector and significantly more likely to work in small firms. Women are more likely to work in temporary

jobs and less likely to work in the formal sector. Both men and women who come of age during a recession delay marriage significantly.

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7 Tables and Figures

Figure 1: National Unemployment Rates

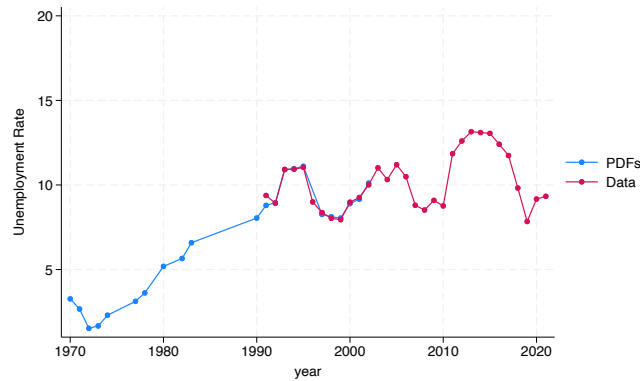
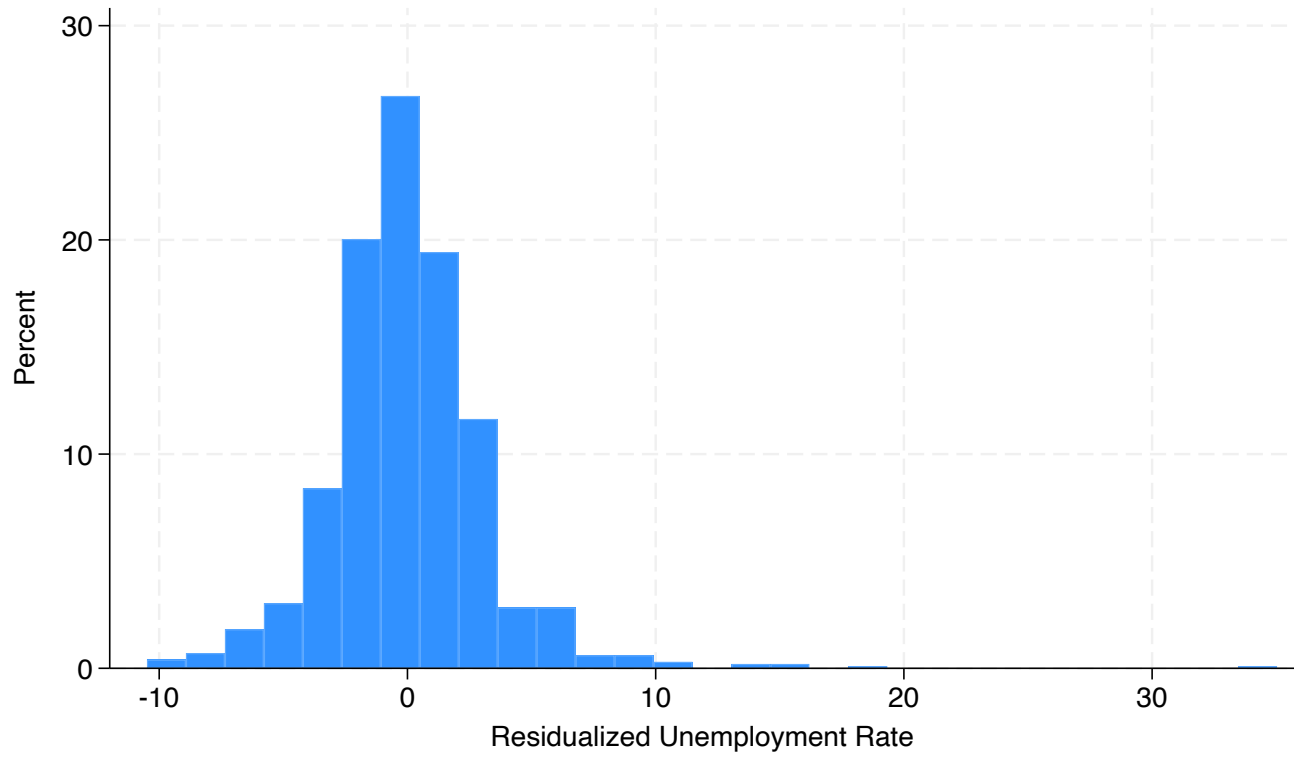


Figure 2: Residualized Governorate Unemployment Rates



Note: This figure reports residualized governorate unemployment rates after including year and governorate fixed effects. The dispersion pictured here ensures adequate identifying variation in the regressions of interest.

Table 1: ELMPS Observation by Round

Round	Freq.
98	6,636
06	4,227
12	4,164
18	22,078
98 & 06	4,143
06 & 12	2,796
12 & 18	16,252
98 & 06 & 12	3,073
06 & 12 & 18	12,756
98 & 06 & 12 & 18	10,145

Table 2: Data quality

	mean	p50	p75	p95	min	max	count
Sex consistent =1	0.99	1	1	1	0	1	49165
YOB consistent =1	0.71	1	1	1	0	1	49165
YOB inconsistencies	0.76	0	1	4	0	81	49164
Age at 1st mar consistent =1	0.38	0	1	1	0	1	18348
Age 1st mar inconsistencies	1.86	1	2	8	0	38	18358
Share conflicting employment status	0.21	0.06	0.32	1.00	0	1	31378
Share conflicting employment status (adjacent)	0.16	0.00	0.25	0.78	0	1	19704

Note: Sex consistent reports whether a person's sex is the same across all surveys. YOB consistent reports whether a person's year of birth is the same across all surveys. YOB inconsistencies reports summary statistics on the range of a given person's reported birthdays. Age at first marriage consistent =1 reports whether a person's year of first marriage is the same across all surveys (conditional on being nonmissing). Age at first marriage inconsistencies reports summary statistics on the range of a given person's reported age at first marriage. Share conflicting employment status reports the share of years in which an individual reported two different employment statuses in different survey waves, restricting to years where the individual was 16 or older. To be conservative, a year is counted as conflicted if reports from at least one survey differed, even if the other 2-3 reports are consistent. The following row reports the same statistic, restricted to consecutive surveys. The final row is the most relevant statistic since the retrospective panel is constructed using the closest survey round.

Table 3: Descriptive Statistics from each ELMPS Wave

	1998		2006		2012		2018	
	Men	Women	Men	Women	Men	Women	Men	Women
<i>Demographics</i>								
Attained Preparatory	0.54	0.34	0.59	0.43	0.63	0.50	0.66	0.54
Attained Secondary	0.47	0.30	0.54	0.39	0.58	0.46	0.61	0.49
Attained Post-Secondary	0.25	0.14	0.25	0.16	0.23	0.18	0.22	0.19
Attained University	0.18	0.09	0.20	0.12	0.19	0.15	0.19	0.16
Employed (past 3 months)	0.88	0.22	0.91	0.28	0.92	0.23	0.90	0.21
Age at First Marriage	26.80	20.08	27.03	20.94	26.80	21.32	26.73	21.39
Observations	12095	11902	18587	18553	24483	24703	30542	30688
<i>Marriage Characteristics and Costs (2010 EGP)</i>								
Husband-wife age gap			6.73	6.78	6.31	6.33	6.19	6.09
Husband-wife age gap			6.93	6.96	6.58	6.61	6.54	6.59
Couple lives alone				0.55	0.65	0.63	0.64	0.63
Consanguineous				0.30	0.30	0.30	0.28	0.28
Polygamous wife				0.01		0.01		0.01
Mahr				5,863	1,772	2,044	2,518	2,699
Deferred Mahr				9,829	6,402	6,915	7,174	8,020
Ayma				9,829	37,999	38,012	48,877	59,823
Jewelry				8,608	6,891	7,202	9,215	8,921
Furniture				38,613	23,697	24,310	24,040	25,687
Gihaz				14,682	12,731	13,518	19,443	18,727
Housing				34,347	27,554	28,854	26,750	37,463
Wedding				6,182	6,207	6,322	6,865	19,763
Total bride side				27,543	13,153	13,508	24,954	28,718
Total groom side				78,509	33,814	32,382	57,733	74,972
Observations			0	6,590	4,654	6,367	4,441	4,840

Note:

Figure 3: National Youth Unemployment Rate

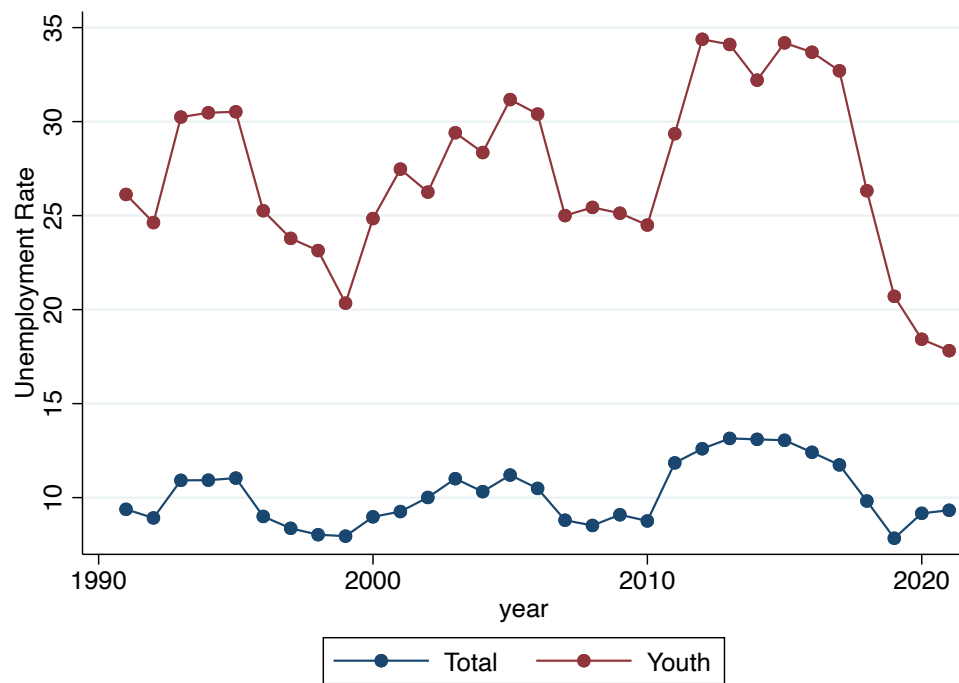


Figure 4: Time from graduation to first job, ELMPS 2018



Table 4: Effect of Entry Conditions on Educational Attainment

	Secondary	Post-secondary	University
<i>Men</i>			
Gov. of Birth, age 18 UR	0.00156 (0.00175)	-0.00194 (0.00191)	-0.000244 (0.00164)
Observations	13092	13092	11051
<i>Women</i>			
Gov. of Birth, age 18 UR	0.00808** (0.00301)	0.000833 (0.00143)	-0.00104 (0.00160)
Observations	12688	12688	10599

Note: This table reports results from a regression of an indicator for having attained at least secondary, post-secondary or university education on the unemployment rate at age 18 using specification 1. There is one observation per individual, including the highest education they report in any round. Individuals aged 25+ at the time their highest observed education is measured are included in the secondary and post-secondary regressions; individuals aged 28+ are included in the university regressions.

Table 5: Long-run Effect of Entry Unemployment Rate on Male Employment

	(1) Employment	(2) WageWork	(3) Self Emp	(4) Self (no employees)	(5) Formal WW
UR*exp=0	-0.00706*** (0.00238)	-0.00562*** (0.00177)	-0.00142 (0.00162)	-0.00247 (0.00149)	-0.000891 (0.00177)
UR*exp=1	-0.00428** (0.00176)	-0.00345** (0.00152)	-0.000818 (0.00129)	-0.00208 (0.00130)	-0.000245 (0.00164)
UR*exp=2	-0.00121 (0.00185)	-0.00184 (0.00159)	0.000623 (0.00124)	-0.000758 (0.00135)	-0.000179 (0.00150)
UR*exp=3	-0.00170 (0.00171)	-0.00274 (0.00163)	0.00101 (0.00114)	-0.000100 (0.00111)	-0.00110 (0.00142)
UR*exp=4	-0.00108 (0.00189)	-0.00250 (0.00182)	0.00140 (0.00102)	0.000655 (0.000984)	-0.00255* (0.00135)
UR*exp=5	0.00134 (0.00159)	-0.000933 (0.00145)	0.00226* (0.00124)	0.00123 (0.00115)	-0.00300** (0.00126)
UR*exp=6	0.00170 (0.00133)	-0.000586 (0.00133)	0.00227** (0.00107)	0.00146 (0.000987)	-0.00300** (0.00110)
UR*exp=7	0.00197 (0.00128)	-0.000679 (0.00136)	0.00264** (0.00111)	0.00152 (0.00100)	-0.00330** (0.00121)
UR*exp=8	0.00226** (0.00107)	-0.000976 (0.00133)	0.00323*** (0.00104)	0.00171 (0.00101)	-0.00348** (0.00135)
UR*exp=9	0.00272** (0.00102)	-0.000647 (0.00125)	0.00337*** (0.00101)	0.00190* (0.00105)	-0.00337** (0.00146)
UR*exp=10+	0.00418*** (0.00144)	0.00118 (0.00130)	0.00302** (0.00130)	0.00290* (0.00163)	-0.00101 (0.00124)
Constant	0.779*** (0.00633)	0.595*** (0.00727)	0.184*** (0.00657)	0.107*** (0.00762)	0.261*** (0.00751)
Current Year	Yes	Yes	Yes	Yes	Yes
Potential Experience	Yes	Yes	Yes	Yes	Yes
Cohort	Yes	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes	Yes
Clustering	Gov Birth	Gov Birth	Gov Birth	Gov Birth	Gov Birth
Entry Yrs	1968-2018 minus 75,76,85-89,96,03				
Obs	305260	304203	304840	304203	304176

Note:

Table 6: Long-run Effect of Entry Unemployment Rate on Female Employment

	(1) Employment	(2) WageWork	(3) Self Emp	(4) Self (no employees)	(5) Formal WW
UR*exp=0	0.00243 (0.00252)	0.000621 (0.00166)	0.00179 (0.00151)	0.00140 (0.00126)	0.00165 (0.00160)
UR*exp=1	0.00249 (0.00255)	0.000789 (0.00168)	0.00168 (0.00150)	0.00131 (0.00127)	0.00160 (0.00163)
UR*exp=2	0.00212 (0.00260)	0.000553 (0.00168)	0.00157 (0.00154)	0.00121 (0.00131)	0.00147 (0.00154)
UR*exp=3	0.00177 (0.00253)	0.000647 (0.00163)	0.00112 (0.00154)	0.000748 (0.00130)	0.00142 (0.00156)
UR*exp=4	0.00112 (0.00260)	0.000231 (0.00163)	0.000888 (0.00154)	0.000560 (0.00133)	0.000794 (0.00158)
UR*exp=5	0.000652 (0.00262)	0.000125 (0.00162)	0.000515 (0.00152)	0.000219 (0.00131)	0.000434 (0.00162)
UR*exp=6	0.000564 (0.00246)	0.000384 (0.00147)	0.000168 (0.00151)	-0.0000964 (0.00131)	0.000337 (0.00156)
UR*exp=7	0.000528 (0.00246)	0.000163 (0.00150)	0.000327 (0.00157)	-0.0000258 (0.00138)	0.000694 (0.00159)
UR*exp=8	0.0000669 (0.00239)	0.0000542 (0.00154)	-0.0000527 (0.00153)	-0.000351 (0.00135)	0.000902 (0.00152)
UR*exp=9	-0.000340 (0.00224)	-0.000134 (0.00144)	-0.000259 (0.00153)	-0.000493 (0.00136)	0.00110 (0.00145)
UR*exp=10+	0.00174 (0.00190)	0.00237 (0.00160)	-0.000656 (0.00124)	-0.000726 (0.00117)	0.00400** (0.00179)
Constant	0.176*** (0.0146)	0.114*** (0.00950)	0.0625*** (0.00907)	0.0600*** (0.00812)	0.0754*** (0.00982)
Current Year	Yes	Yes	Yes	Yes	Yes
Potential Experience	Yes	Yes	Yes	Yes	Yes
Cohort	Yes	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes	Yes
Clustering	Gov Birth	Gov Birth	Gov Birth	Gov Birth	Gov Birth
Entry Yrs	1968-2018 minus 75,76,85-89,96,03				
Obs	312079	311093	311866	311093	311088

Table 7: Effect of Entry Conditions on Earnings and Wages

	Earnings < p50	Earnings < p70	Earnings < p90	Log Wages
<i>Men, all</i>				
Gov. of Birth, age 18 UR	0.00318** (0.00114)	0.00209 (0.00136)	0.00114 (0.000747)	0.0000338 (0.00243)
Observations	25472	25472	25472	18242
<i>Men, < secondary></i>				
Gov. of Birth, age 18 UR	0.00476 (0.00288)	0.000182 (0.00210)	0.00136 (0.000933)	-0.000185 (0.00364)
Observations	8322	8322	8322	6487
<i>Men, secondary</i>				
Gov. of Birth, age 18 UR	-0.00140 (0.00189)	0.000307 (0.00206)	-0.000944 (0.00120)	0.00281 (0.00374)
Observations	10524	10524	10524	7234
<i>Men, university</i>				
Gov. of Birth, age 22 UR	0.00441** (0.00183)	0.00873*** (0.00205)	0.00645*** (0.00226)	-0.00556 (0.00573)
Observations	6091	6091	6091	4051
<i>Women, all</i>				
Gov. of Birth, age 18 UR	0 (.)	0 (.)	-0.0000462 (0.00116)	-0.00522 (0.00543)
Observations	31310	31310	31310	3914

Note: This table reports results from a regression of an indicator ... on the unemployment rate at age 18 using specification 1.

Table 8: Hazard into Marriage

	Men		Women	
Gov. of Birth, age 18 UR	0.985*** (0.00451)		0.989** (0.00507)	
Gov. of Birth, age 12 UR		0.991 (0.00774)		0.987** (0.00571)
Cohort	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes
Clustering				
Obs	19859	20455	20110	20700

Note: This figure reports hazard ratios from a Cox proportional hazards model on the effect of unemployment rate at entry on the time to marriage. A one point increase in the unemployment rate at entry reduces men's probability of getting married each year by 1.1 percent.

Table 9: Effect of Recession Entry on Marriage Outcomes

	Men			Women, UR at 18			Women, UR at 12		
	Effect	Mean	N	Effect	Mean	N	Effect	Mean	N
Husband-wife age gap	0.00 0.03	6.67	6,445	-0.04 0.03	6.37	7,785	-0.06* 0.03	6.37	6,880
Husband-wife age gap	0.00 0.03	6.91	6,445	-0.03 0.03	6.66	7,785	-0.05 0.03	6.66	6,880
Couple lives alone	0.00 0.01	0.62	4,663	0.00 0.00	0.59	7,677	0.00 0.00	0.59	7,406
Consanguineous	0.00 0.00	0.30	4,663	0.00 0.00	0.30	7,677	0.00 0.00	0.29	7,406
Mahr	73.90 106.42	1,973	2,467	100.25 139.29	3,894	5,560	-123.69 273.85	4,760	5,121
Mahr Muakhr	162.60 183.64	7,215	2,519	124.06 151.16	8,683	5,950	-191.17 259.56	9,299	5,594
Ayma	14.71 616.91	40,359	2,649	-335.65 213.01	40,070	4,262	351.30 354.28	39,924	3,518
Jewelry	156.90 168.90	7,919	2,785	-13.84 100.82	8,129	6,320	-232.27 193.80	8,633	5,950
Furniture	24.20 295.37	26,381	2,610	579.30* 305.56	32,958	6,139	1,051.59** 447.83	35,323	5,801
Gihaz	436.66 288.39	15,059	2,510	-109.35 129.76	14,761	6,152	16.23 185.85	15,624	5,814
Housing	-497.14 1,108.89	30,658	2,465	918.55 593.29	31,192	5,909	479.77 1,081.21	32,912	5,607
Wedding	46.06 199.84	6,937	2,634	17.77 72.62	6,126	6,150	-10.43 150.13	6,587	5,814
Total bride side	127.80 231.72	16,243	2,729	243.19 292.32	23,092	6,252	237.71 545.46	24,739	5,896
Total groom side	672.96 1,093.90	40,709	2,811	1,386.57* 794.13	60,508	6,363	1,035.14 1,446.44	66,374	5,995

Note: This table reports estimates from 28 separate regressions of the effect of unemployment rate at entry of various marriage outcomes. The columns marked men (women) use data reported by men (women) pertaining to their marriages and the unemployment rate in the individual's governorate of birth, when the individual was aged 18. To avoid censoring bias, I restrict to individuals observed at an age greater than or equal to the 95th percentile of age at first marriage (33 for men, 29 for women). Husband-wife age gap is the difference between the husband and the wife's age (so older husband is a positive number). The following row is the absolute value of this quantity. Couple lives alone is an indicator equal to one if the couple lives in a home or apartment without other family in the house. Consanguineous is an indicator equal to one if the couple are related by blood (ie, cousins). Mahr is obligatory under Islam and refers to the payment made from the husband to the wife at the time of marriage. Mahr Mukahr is the deferred mahr owed to the wife in the event of divorce (except wife-initiated no-fault divorce) or her husband's death. Ayma refers to a non-religious but culturally customary contract detailing the goods/wealth brought into the marriage by the wife which may be reclaimed by the wife under certain circumstances. Jewelry refers to the value of gold jewelry gifted to the bride by the groom's and his family. Furniture refer to the value of home furnishings purchased for the marital home. Gihaz refers to other preparations for the home (ie, linens, kitchenware, etc). Housing refers to cost of housing incurred for the couple's marital home, and wedding is the cost of the wedding itself. Total bride side sums the amount the bride or her family spent on the furniture, gihaz, housing and wedding. Total groom side sums their groom's side's share of the same expenses, plus the mahr and jewelry. (Mahr muakhr and ayma are not included in this sum since they are only due upon divorce/death.)

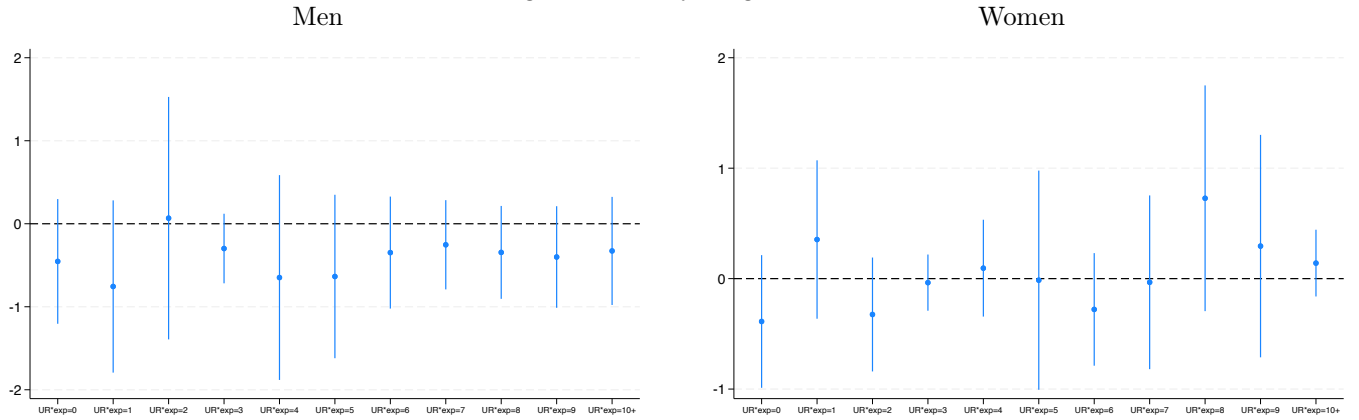
Table 10: Marital Decisionmaking and Women's Mobility

	Men	Women			
	Decision	Decision	Mobility	Decision	Mobility
Gov. of Birth, age 18 UR	0.00348 (0.00513)	0.00270 (0.00301)	-0.00313 (0.00451)		
Gov. of Birth, age 12 UR				0.0106** (0.00487)	-0.00167 (0.00455)
Cohort	Yes	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes	Yes
Clustering	GB	GB	GB	GB	GB
Obs	5234	12642	12624	11046	11030

Note: This table reports the effect of entry conditions (or entry conditions of women's likely marriage cohort) on marital decisionmaking and women's mobility. The decision columns report impacts on an index of marital decisionmaking, where an increase corresponds to an increase in joint or wife-only decisionmaking.

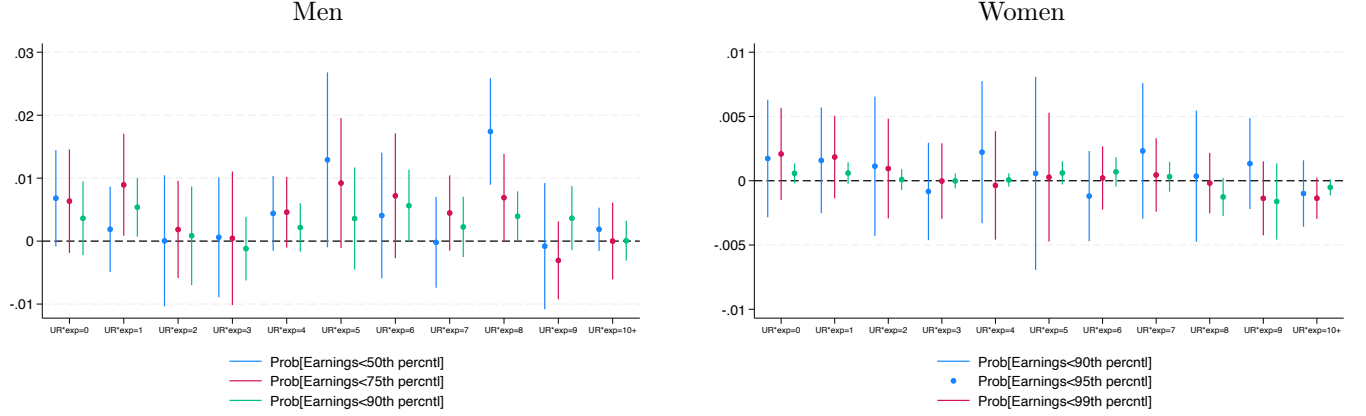
8 Appendix Tables and Figures

Figure 5: Hourly Wages

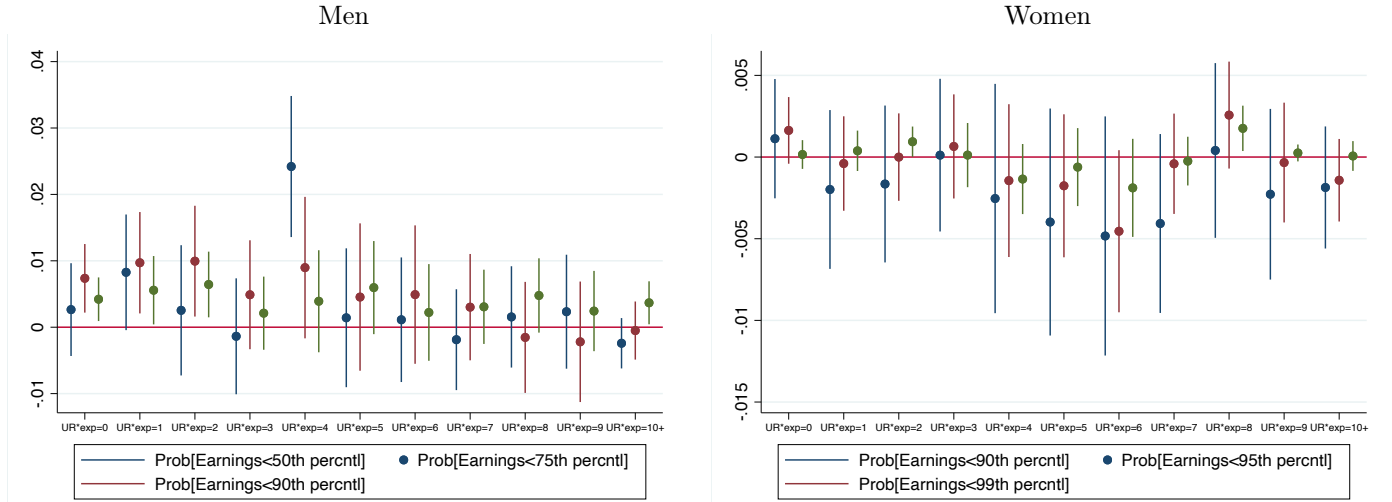


Note: This figure reports coefficients from a regression of hourly wages on unemployment rate at entry using specification 2. The sample includes wage observations from panel surveys only, not retrospective job histories. The median wage is 10 egyptian pounds for men and women. The mean male wage is 13.9 and the mean female wage is 13.5.

Figure 6: Effect on Earnings CDFs
Assuming entry at age 18



Assuming entry at age 22



Note: These figures report coefficients from a regression of an indicator for having earnings $\leq$ the n th percentile on the unemployment rate at entry. The first two figures use the unemployment rate in the governorate of birth at age 18 and only include employment observations for individuals aged 18 and above. The second two figures use the unemployment rate in the governorate of birth at age 22 and only include employment observations for individuals aged 22 and above.

Table 11: Long-run Effect of Entry Unemployment Rate on Male Employment: Less than Secondary

	(1) Employment	(2) Self Emp	(3) WageWork	(4) Formal WW	(5) Public Entprse Formal WW
UR*exp=0	-0.00801 (0.00522)	-0.00407 (0.00273)	-0.00394 (0.00395)	0.00128 (0.00213)	0.00117 (0.000921)
UR*exp=1	-0.00394 (0.00396)	-0.00271 (0.00245)	-0.00122 (0.00345)	0.00221 (0.00208)	0.00162 (0.000994)
UR*exp=2	0.000267 (0.00377)	-0.00136 (0.00241)	0.00163 (0.00330)	0.00374* (0.00217)	0.00225* (0.00114)
UR*exp=3	-0.000210 (0.00399)	-0.00109 (0.00234)	0.000877 (0.00362)	0.00436* (0.00217)	0.00259** (0.00116)
UR*exp=4	0.00102 (0.00390)	-0.000847 (0.00227)	0.00185 (0.00348)	0.00477** (0.00209)	0.00270** (0.00118)
UR*exp=5	0.00262 (0.00359)	-0.000392 (0.00237)	0.00302 (0.00311)	0.00474** (0.00190)	0.00258* (0.00127)
UR*exp=6	0.00291 (0.00325)	-0.000415 (0.00224)	0.00335 (0.00293)	0.00443** (0.00181)	0.00233* (0.00130)
UR*exp=7	0.00340 (0.00288)	-0.000989 (0.00250)	0.00440 (0.00293)	0.00500** (0.00188)	0.00257* (0.00142)
UR*exp=8	0.00334 (0.00273)	-0.000252 (0.00234)	0.00362 (0.00274)	0.00516** (0.00193)	0.00266* (0.00147)
UR*exp=9	0.00374 (0.00286)	0.000342 (0.00257)	0.00342 (0.00286)	0.00551*** (0.00189)	0.00269* (0.00152)
UR*exp=10+	0.00232 (0.00234)	0.000967 (0.00281)	0.00142 (0.00261)	0.00591** (0.00225)	0.00404* (0.00198)
Constant	0.849*** (0.0160)	0.284*** (0.0129)	0.567*** (0.0141)	0.128*** (0.0107)	0.0141 (0.00862)
Current Year	Yes	Yes	Yes	Yes	Yes
Potential Experience	Yes	Yes	Yes	Yes	Yes
Cohort	Yes	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes	Yes
Clustering	Gov Birth	Gov Birth	Gov Birth	Gov Birth	Gov Birth
Entry Yrs	1968-2018 minus 75,76,85-89,96,03				
Obs	129157	128986	128526	128518	128509

Note:

Table 12: Long-run Effect of Entry Unemployment Rate on Male Employment: Secondary Grads

	(1) Employment	(2) Self Emp	(3) WageWork	(4) Formal WW
UR*exp=0	-0.00801 (0.00522)	0.0000709 (0.00195)	-0.000552 (0.00200)	-0.000301 (0.00220)
UR*exp=1	-0.00394 (0.00396)	-0.000692 (0.00226)	0.00107 (0.00233)	0.00106 (0.00216)
UR*exp=2	0.000267 (0.00377)	0.000168 (0.00208)	0.000712 (0.00249)	0.000330 (0.00220)
UR*exp=3	-0.000210 (0.00399)	0.000630 (0.00192)	-0.00168 (0.00246)	-0.00162 (0.00215)
UR*exp=4	0.00102 (0.00390)	0.000383 (0.00201)	-0.00269 (0.00240)	-0.00375 (0.00227)
UR*exp=5	0.00262 (0.00359)	0.00139 (0.00209)	-0.00220 (0.00226)	-0.00498** (0.00225)
UR*exp=6	0.00291 (0.00325)	0.000976 (0.00219)	-0.00265 (0.00213)	-0.00547** (0.00229)
UR*exp=7	0.00340 (0.00288)	0.00169 (0.00230)	-0.00298 (0.00243)	-0.00651** (0.00250)
UR*exp=8	0.00334 (0.00273)	0.00168 (0.00265)	-0.00289 (0.00266)	-0.00747** (0.00276)
UR*exp=9	0.00374 (0.00286)	0.00168 (0.00278)	-0.00184 (0.00283)	-0.00736** (0.00297)
UR*exp=10+	0.00232 (0.00234)	0.00163 (0.00374)	0.00237 (0.00291)	-0.00597* (0.00344)
Constant	0.849*** (0.0160)	0.154*** (0.0214)	0.621*** (0.0173)	0.293*** (0.0199)
Current Year	Yes	Yes	Yes	Yes
Potential Experience	Yes	Yes	Yes	Yes
Cohort	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes
Clustering	Gov Birth	Gov Birth	Gov Birth	Gov Birth
Entry Yrs	1968-2018 minus 75,76,85-89,96,03			
Obs	129157	102556	102492	102474

Note:

Table 13: Long-run Effect of Entry Unemployment Rate on Male Employment: University Grads

	(1) Employment	(2) Self Emp	(3) Employer	(4) Self (No emp)	(5) WageWork	(6) Formal WW
UR*exp22=0	-0.00107 (0.00247)	-0.00245 (0.00205)	-0.000669 (0.00103)	-0.00178 (0.00152)	0.00126 (0.00290)	0.00499* (0.00280)
UR*exp22=1	0.000373 (0.00270)	-0.00116 (0.00260)	-0.000570 (0.00100)	-0.000598 (0.00221)	0.00132 (0.00292)	0.00342 (0.00269)
UR*exp22=2	0.00225 (0.00264)	0.000181 (0.00250)	-0.000855 (0.000945)	0.00103 (0.00222)	0.00194 (0.00316)	0.00116 (0.00289)
UR*exp22=3	-0.000679 (0.00295)	0.00119 (0.00252)	-0.000198 (0.00105)	0.00139 (0.00202)	-0.00195 (0.00300)	-0.00366 (0.00286)
UR*exp22=4	-0.00336 (0.00274)	0.00175 (0.00229)	-0.000270 (0.00108)	0.00202 (0.00183)	-0.00512 (0.00301)	-0.00517 (0.00314)
UR*exp22=5	-0.00209 (0.00228)	0.00239 (0.00235)	0.000444 (0.00124)	0.00195 (0.00179)	-0.00446 (0.00291)	-0.00622* (0.00329)
UR*exp22=6	-0.00310 (0.00230)	0.00174 (0.00216)	0.0000925 (0.00139)	0.00166 (0.00165)	-0.00479* (0.00232)	-0.00675** (0.00309)
UR*exp22=7	-0.00303 (0.00193)	0.00166 (0.00212)	0.000602 (0.00147)	0.00107 (0.00143)	-0.00465* (0.00236)	-0.00743** (0.00334)
UR*exp22=8	-0.00232 (0.00151)	0.00215 (0.00216)	0.00150 (0.00172)	0.000658 (0.00127)	-0.00446** (0.00211)	-0.00576 (0.00338)
UR*exp22=9	-0.00313** (0.00150)	0.00278 (0.00213)	0.00175 (0.00164)	0.00103 (0.00141)	-0.00589*** (0.00193)	-0.00735** (0.00311)
UR*exp22=10+	-0.000940 (0.00150)	0.00511* (0.00257)	0.00281 (0.00254)	0.00231 (0.00182)	-0.00591* (0.00286)	-0.00853** (0.00325)
Constant	0.816*** (0.0114)	0.0907*** (0.0142)	0.0469*** (0.0109)	0.0442*** (0.0122)	0.730*** (0.0140)	0.561*** (0.0183)
Current Year	Yes	Yes	Yes	Yes	Yes	Yes
Potential Experience	Yes	Yes	Yes	Yes	Yes	Yes
Cohort	Yes	Yes	Yes	Yes	Yes	Yes
Gov of birth	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Gov Birth	Gov Birth	Gov Birth	Gov Birth	Gov Birth	Gov Birth
Entry Yrs	1968-2018 minus 75,76,85-89,96,03					
Obs	54369	54336	53947	53947	53947	53947

Note: